

Practical : 11 : Write a PHP program to perform search operation on the student records using AJAX

Code:

Index.php :

```
<?php
// No PHP logic needed in this file; it's for HTML, CSS, and JavaScript
?>

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Search</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      display: flex;
      justify-content: center;
      align-items: center;
      min-height: 100vh;
      margin: 0;
      background-color: #f0f0f0;
    }
    .container {
      background-color: white;
      padding: 20px;
      border-radius: 8px;
      box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
      width: 90%;
    }
  </style>
</head>
<body>
  <div class="container">
    <h2>Student Search</h2>
    <form>
      <input type="text" value="Search Student Record" />
      <input type="button" value="Search" />
    </form>
  </div>
</body>
</html>
```

```
    max-width: 600px;
}
h2 {
    text-align: center;
    color: #333;
    margin-bottom: 20px;
}
.search-group {
    margin-bottom: 20px;
}
input[type="text"] {
    width: 100%;
    padding: 10px;
    font-size: 16px;
    border: 1px solid #ccc;
    border-radius: 4px;
    box-sizing: border-box;
}
table {
    width: 100%;
    border-collapse: collapse;
    margin-top: 10px;
    display: none;
}
th, td {
    padding: 10px;
    text-align: left;
    border-bottom: 1px solid #ddd;
}
th {
    background-color: #4CAF50;
```

```
        color: white;
    }
    tr:nth-child(even) {
        background-color: #f9f9f9;
    }
    .error {
        color: red;
        font-size: 14px;
        margin-top: 10px;
        text-align: center;
        display: none;
    }
</style>
</head>
<body>
    <div class="container">
        <h2>Search Student Records</h2>
        <div class="search-group">
            <input type="text" id="searchInput" placeholder="Enter student name or ID..."
onkeyup="searchStudents()">
        </div>
        <div id="error" class="error"></div>
        <table id="resultsTable">
            <thead>
                <tr>
                    <th>ID</th>
                    <th>Name</th>
                    <th>Grade</th>
                </tr>
            </thead>
            <tbody id="resultsBody"></tbody>
        </table>
    </div>
</body>
</html>
```

```

</table>

</div>

<script>

function searchStudents() {

    const query = document.getElementById('searchInput').value.trim();

    const resultsTable = document.getElementById('resultsTable');

    const resultsBody = document.getElementById('resultsBody');

    const errorDiv = document.getElementById('error');


    // Clear previous results and errors

    resultsBody.innerHTML = '';

    errorDiv.style.display = 'none';

    resultsTable.style.display = 'none';


    if (query.length === 0) {

        return; // No query, no request

    }


    // Create AJAX request

    const xhr = new XMLHttpRequest();

    xhr.open('POST', 'search.php', true);

    xhr.setRequestHeader('Content-Type', 'application/x-www-form-urlencoded');


    xhr.onreadystatechange = function () {

        if (xhr.readyState === 4) {

            if (xhr.status === 200) {

                const response = JSON.parse(xhr.responseText);

                if (response.error) {

                    errorDiv.textContent = response.error;

                    errorDiv.style.display = 'block';

                }

            }

        }

    }

}

```

```

    } else if (response.students.length > 0) {
        response.students.forEach(student => {
            const row = document.createElement('tr');
            row.innerHTML = `
                <td>${student.id}</td>
                <td>${student.name}</td>
                <td>${student.grade}</td>
            `;
            resultsBody.appendChild(row);
        });
        resultsTable.style.display = 'table';
    } else {
        errorDiv.textContent = 'No students found';
        errorDiv.style.display = 'block';
    }
} else {
    errorDiv.textContent = 'Error connecting to server';
    errorDiv.style.display = 'block';
}
}
};

// Send query to PHP script
xhr.send('query=' + encodeURIComponent(query));
}
</script>
</body>
</html>

```

Search.php :

```
<?php

// Simulated student records (no database, stored in array)

$students = [

    ['id' => 'S001', 'name' => 'Roshan', 'grade' => 'A'],
    ['id' => 'S002', 'name' => 'Pranjal', 'grade' => 'B+'],
    ['id' => 'S003', 'name' => 'Arjun', 'grade' => 'A+'],
    ['id' => 'S004', 'name' => 'Bobby', 'grade' => 'B'],
    ['id' => 'S005', 'name' => 'Ronni', 'grade' => 'A++'],
    ['id' => 'S006', 'name' => 'puppy', 'grade' => 'A'],
    ['id' => 'S007', 'name' => 'Jhonny deph', 'grade' => 'C'],
    ['id' => 'S008', 'name' => 'Karan aujla', 'grade' => 'A+'],
    ['id' => 'S009', 'name' => 'Emma watson', 'grade' => 'B++'],

];

// Get query from AJAX request

$query = isset($_POST['query']) ? trim($_POST['query']) : "";

$response = ['students' => [], 'error' => ""];

if (empty($query)) {

    $response['error'] = 'Search query cannot be empty';

} else {

    // Search students by ID or name (case-insensitive)

    foreach ($students as $student) {

        if (

            strpos($student['id'], $query) !== false ||

            strpos($student['name'], $query) !== false

        ) {

            $response['students'][] = $student;

        }

    }

}
```

```
}
```

```
// Return JSON response
```

```
header('Content-Type: application/json');
```

```
echo json_encode($response);
```

```
?>
```

Search Student Records

ID	Name	Grade
S002	Pranjal	B+
S006	puppy	A
S007	Jhonny deph	C

Search Student Records

ID	Name	Grade
S001	Roshan	A
S002	Pranjal	B+
S003	Arjun	A+
S005	Ronni	A++
S008	Karan aujla	A+